

Pratham M

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INTRODUCTION

Software Engineer with hands-on experience across backend development, DevOps, deployment, C#, .NET, Angular, Kubernetes, Docker, Industrial IoT, AI automation, Python, AI, MCP, and system validation. Strong ownership and leadership mindset, with a proven ability to work across technical streams, quickly learn new technologies, and deliver reliable end-to-end solutions.

EXPERIENCE

Trivium eSolution Pvt Ltd [January 2025 -Present]

Bengaluru, Karnataka

Associate Software Engineer

October 2025 – Present

- **Transformer-Digitalization – Siemens Energy:** Migrated software components from AMD64 to ARM64 architecture and validated deployment on Siemens IOT2050 industrial edge device running Debian Linux.
- Set up and deployed the application on a **Siemens IOT2050 ARM64 industrial IOT device** by connecting the embedded ARM board via **UART and DC power supply**; configured **Debian 12**, built ARM64 artifacts, and validated the project end-to-end on the target .
- Build the TMW Binaries via GitLab and replaced the Libraries of amd64 with the ARM64 architecture so the artifact could build on IOT2050 device.
- Contributed to backend and DevOps enhancements by implementing OData sorting/filtering improvements, resolving YAML and K3s deployment issues, enabling ARM build configurations, improving upgrade deployment workflows, and fixing Keycloak authentication/redirection issues using Kubernetes, Docker, Bash scripting, and YAML code improvement.
- Refactored backend services using **.NET** (ConfigurationServiceWorker, AuthWorker,EventWorker) and rebuilt frontend with **Angular**; implemented **configuration-driven architecture** for dynamic UI generation.
- Authored **PowerShell automation scripts** to build Docker artifacts and deploy microservices to **Kubernetes (K3s)**; configured **Keycloak, Kong API Gateway, RabbitMQ, and PostgreSQL**.
- Integrated real-time **Modbus Python** data pipeline for live visualization of **DGA Moisture, DGA ROC, SCADA**, and Trend Charts; validated UI stability with **5,000+ row datasets**.
- Developed and validated Linux shell scripts across Debian 12/13 and industrial Linux environments for deployment automation and system configuration.; enabled **multi-language** support for dynamic UI translation.
- Increased backend test coverage from **55% to 85%** by implementing comprehensive **xUnit unit tests** and addressing **SonarQube code quality issues**.
- Have worked on writing Kubernetes scripts yamls for the production services
- Have worked on writing Dockerfile for Keycloak Theme integration for docker-compose and K3s-airgap deployment
- worked on fixing the keycloak pods when there were going to CrashLoop on every update of artifact
- Independently validated end-to-end backend and frontend functionality on IPC systems (Debian 12 OS) and WSL environments; tested Bash scripts across Debian 12/13, Industrial OS, and Oracle VM, and verified ARM64 artifacts on the Siemens IOT2050 edge device to ensure cross-platform stability before production rollout.
- wrote Kubernetes scripts and kubectl configurations embedded in deployment artifacts; set up symlinks to ensure deployment scripts, K3s installation, and uninstallation workflows execute correctly across environments.
- Have tested the System via ModSim, IEDScount, IEDExplorer, OPCUA, Python Scripts
- Connected MCP Server via postgres to access all the tables and database information by writing an markdown file , Connected an playwright MCP so that it would help for testing all the UI component via Github Copilot.

Trainee

July 2025 – September 2025

- **Daily Data Warehouse:** Built web apps using **React** and **.NET**; developed reusable UI components with **styled-components**, implemented Badge filtering, documented via **Storybook**, and designed interactive **D3.js** charts with hover tooltips.
- **Model Context Protocol:** Built a personal automation system using **Claude Desktop** with a custom MCP to connect **Gmail, GitHub, and Fi app**; automated email workflows, natural-language GitHub operations, and real-time finance data analysis.

Intern

January 2025 – June 2025

- Built **AI Security & Vulnerability Awareness Tool** using **OpenAI API and Ollama** to extract CVSS, CWE, CVE data from pentest PDFs, reducing manual analysis by **80%**.
- Developed full-stack **E-Commerce** app (Angular + Spring Boot + MySQL + Kafka) with SMTP OTP auth, admin dashboard, and inventory management; implemented **DevSecOps CI/CD** with Jenkins, SonarQube, Prometheus, and Grafana.

EDUCATION

CHRIST (Deemed to be University)

Bengaluru, Karnataka

Master of Computer Applications – 68.61%

July 2023 – April 2025

St. Joseph's College (Autonomous)

Bengaluru, Karnataka

B.Sc. Computer Science, Mathematics & Electronics – 70.35%

August 2020 – May 2023

Jnanodaya Pre-University College

Bengaluru, Karnataka

Electronics, Mathematics, Computer Science – 79.00%

June 2018 – April 2020

TECHNICAL SKILLS

Languages & Frameworks: C ,Python, C#/.NET, Java, Spring Boot, Angular, React, Golang, Bash/Shell, PowerShell, HTML, JavaScript

DevOps & Cloud: Docker, Kubernetes (K3s/K8s), Jenkins, SonarQube, Prometheus, Grafana, AWS (EC2, IAM), Azure

Databases & Messaging: PostgreSQL, MySQL, Firebase, RabbitMQ, Kafka

Data Engineering Skills: Data Analytics & Reporting, Tableau Dashboard Development, Python (pandas, numpy, matplotlib), Data Visualization & Storytelling, Automation & Workflow Optimization, Supply Chain Analytics (Basic), Large Language Models (LLMs) for Automation, Stakeholder Collaboration & Communication, Git & Version Control, Exploratory Data Analysis (EDA)

Tools & Platforms: Keycloak, Kong API Gateway, Modbus, Git,,GitLab GitHub, Jira, Tableau, Storybook, D3.js

RESEARCH PAPERS, CERTIFICATIONS & CONFERENCES

IEEE Paper: “Live Emotion-Based Music Recommender System” – IITCEE 2026, BNMIT

- Presented at the **4th IEEE International Conference on Intelligent & Innovative Technologies in Computing, Electrical & Electronics**. [[View On IEEE Page](#)] [[View Certificate](#)]

Paper: “Data Security in Cloud Computing” – March 2024

- [[View Paper](#)]

Certifications: AWS Academy Cloud Foundations | Machine Learning (Coursera) | DevOps CI/CD with Git, Jenkins, Ansible & Kubernetes (Udemy) | Android App Development (CHRIST University).

Conferences: AgentCon Bengaluru – Global AI Community, June 20, 2026 | MCP Community Connect – September 26, 2026.